

PREKSHA PRACHEE BHARADWAJ

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Summary

Software Engineer with hands-on experience in Python, SQL, and AI/ML. Passionate about solving real-world problems through data-driven solutions and machine learning. Completed multiple production-grade ML projects – brain tumor segmentation, sleep apnea detection, and PPE compliance detection – with **85.6–95%+** accuracy, and a proven track record of shipping features that measurably reduce risk, latency, and manual effort.

Education

VIT Bhopal

B.Tech in Computer Science (AI & ML) – CGPA: 8.25/10

2023 – 2027

Bhopal, India

Projects

Brain Tumor Segmentation | Python, TensorFlow, Keras, OpenCV, NumPy, Scikit-learn

Aug 2024

- Developed a deep learning segmentation pipeline using U-Net architecture on MRI datasets, achieving **95%+** accuracy and reducing manual radiologist review time by **60% (20 min → 8 min per scan)**
- Optimized inference speed by **75% (8s → 2s per image)** through model pruning and batch normalization, enabling near real-time clinical deployment
- Generated estimated ROI of **\$15,000+** annually by enabling **12 additional scans/day** at same staffing levels; published findings contributing to early-stage cancer detection research

PPE Compliance Detection | Python, YOLOv8, OpenCV, NumPy

Jun 2024

- Developed a real-time YOLOv8 computer-vision system detecting PPE non-compliance with **85%** accuracy, targeting a safety gap linked to **30–40%** of **60,000+** annual construction fatalities globally
- Automated site safety audits across **50+** cameras and **15+** construction sites through distributed inference pipeline and alerting system
- Reduced manual inspection time by **85% (4 hours → 35 min per site)**, generating estimated ROI of **\$50,000+** in avoided incidents and regulatory fines

Sleep Apnea Detection | Python, Scikit-learn, TensorFlow, SciPy, MATLAB

Mar 2025

- Built a multi-modal ML classification system fusing EEG, ECG, and respiratory signals to address **80%** accuracy limitation of single-signal models, achieving **92%** accuracy (**15%** improvement)
- Engineered a real-time detection prototype with **5-second** response windows by implementing sliding-window feature extraction and lightweight ensemble classifiers
- Cut diagnostic latency by **85%** vs standard overnight polysomnography review, with potential ROI of **\$20,000+** in reduced monitoring costs and **40%** faster patient throughput

Technical Skills

Languages: C++, Python, Java, SQL

Frameworks and Libraries: NumPy, Pandas, Matplotlib, Scikit-learn, Langchain

Database & Caching: SQL

ML / CV: TensorFlow, Keras, Scikit-learn, YOLOv8, PyTorch

Coursework: DSA, OS, OOP, DBMS

Tools & Cloud: Git, GitHub, AWS

Experience

Elevance Skill

Feb 2026 – Mar 2026

Full Stack Developer

- Helped contribute to an AI-powered meal recommendation system for Indian diabetes patients, working on logic to suggest regionally relevant, glycemic-index-aware meal plans
- Collaborated on the recommendation backend, integrating patient dietary preferences and blood-sugar trends, and helped test the logic to improve suggestion relevance for everyday Indian diets

Certifications

AWS Cloud Practitioner (AWS) | Applied ML in Python (Coursera) | Anthropic Skill Jar Certification (Anthropic)

Extracurriculars

Cyber for Her Hackathon Finalist - Selected as finalist, demonstrating excellence in cybersecurity and problem-solving

Core Content Writer (Freelancing Club) - Managed editorial workflows and content strategy, increasing club engagement by **40%**

Head Girl - Coordinated student activities and represented student body, managing **25+** student leaders